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Leveraging the power of AI-enabled databases

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Q So how does Oracle's Autonomous Database and Oracle Cloud differentiate itself from the competition with regards to compute and cost perspective?

Agarwal: So we believe AI is very important but there aren't that many data scientists and machine learning experts out there. So our goal has been to embed intelligence into each of our cloud services, especially the Autonomous database, and also in the other parts of the portfolio. With the Autonomous Database, our focus has been to reduce the cost of operations. DBAs have to spend a lot of time towards provisioning databases, scaling databases, patching databases, etc., so our goal here is to have a database that is self-healing, self-patching, self-provisioning. And so we've taken all the knowledge that we've built up and we've learnt about workload management, memory management, etc., to state that DBAs shouldn't have to set indexes nor should they have to figure out how to optimise queries. We should be doing that ourselves. So with the Autonomous Data Warehouse, we're

Oracle helps developers worry less about managing databases and focus more on developing products

turning off those capabilities and we're merging that along with the engineered systems. We've done a lot to optimise performance on the engineered systems. So this Autonomous Data Warehouse is leveraging all our learnings and is running on our Autonomous Database and is relying on the Oracle Cloud for the operations. This way, the DBAs don't have to think about those tasks and they can think about higher-level tasks. Then there is the autonomous transaction processing database which is good for any workloads. There we give more flexibility for DBAs to optimise indexes, set indexes, etc. There again, the patching and self-driving notion of it all comes into play.

Q During my interactions with developers, I've noticed that practically everyone learns about databases using Oracle right from college. However, once they start working on actual products, they tend to go with free alternatives such as MySQL, MongoDB, etc. Is Oracle doing anything to make it easier for developers to start off with Oracle rather than migrating much later into the dev cycle?

Agarwal: That's exactly one of the problems that the Autonomous Database addresses, because now the developer doesn't have to think about installing an Oracle database or setting SGA pools or defining indexes. These tasks would have needed a DBA. All of that is now taken care of by us. They just want a datastore that scales as their business scales. A big focus for us has been to make sure that the developers are now able to connect to a data store without having to worry about scaling, patching or managing the database from a DBA perspective.

Q Oracle is generally thought of as an on-prem solution and developers increasingly prefer cloud solutions. So what's being done to change this perception of Oracle?

Agarwal: We're doing a tonne of events across the world, and if you go to any of our keynotes and events, they're all based on the ability to be able to get a database cloud service. We're making sure that developers and IT organisations understand that they can easily move their databases from on-prem to the public cloud. The way we're doing that is, firstly, it's the same database. The one that was running on prem is running on the cloud. We just added the intelligence and the cloud capabilities around the databases so that you can consume it as a service. When you look at other providers, their databases don't run anywhere except on the public cloud. So making sure that we have the ability to run the same database in both environments means the tools and applications continue working. Secondly, we are giving the developers many different ways to move the database that is on-prem, to the public cloud. For example, you can do an RMAN-based backup of your data and you can provision a cloud-based instance with that backup, but now, that's being delivered as a service. 